

NORGAN TABLET 400mg

NAME AND STRENGTH OF ACTIVE INGREDIENT(s):

Each tablet contains:

Norfloxacin.....400mg

PRODUCT DESCRIPTION:

A white, scored, oblong tablet of size 17x6.5mm with marking 'LIFE'.

PHARMACODYNAMICS:

Norfloxacin is bactericidal and acts intracellularly by inhibiting the A subunit of DNA gyrase, a type II topoisomerase essential for ATP-dependent coiling and supercoiling of bacterial DNA, enabling it to be replicated and repacked in both daughter cells, inhibits relaxation of supercoiled DNA; and promotes double-stranded DNA breakage.

PHARMACOKINETICS:

Peak plasma concentrations of about 1.5µg of norfloxacin per ml have been achieved 1 to 2 hours after the administration of a 400mg dose by mouth; the presence of food can delay absorption. The plasma half-life is about 4 hours and may be prolonged in renal failure; a value of 6.5 hours or more has been reported when creatinine clearance is below 30ml per minute. Norfloxacin has been reported to be 14% bound to plasma proteins. It is probably widely distributed, but information is limited. Norfloxacin penetrates well into tissues of the genito-urinary tract. It crosses the placenta. Relatively high concentrations are achieved in bile. About 30% of a dose is excreted unchanged in the urine within 24 hours producing high urinary concentrations; norfloxacin is least soluble at a urinary pH of 7.5. Urinary excretion is by tubular secretion and glomerular filtration and is reduced by probenecid although plasma concentrations of norfloxacin are not generally affected. Some metabolism occurs, possibly in the liver, and several metabolites have been identified in urine, some with antibacterial activity. About 30% of an oral dose is recovered from the faeces.

INDICATION:

Norfloxacin is used mainly in the treatment of urinary-tract infections. It is indicated in adults in the treatment of complicated and uncomplicated urinary tract infections caused by *Citrobacter freundii*, *Enterobacter cloacae* (*Aerobacter cloacae*), *Escherichia coli*, *Klebsiella pneumoniae*, *Proteus mirabilis*, indole-positive *Proteus* species (including *P. vulgaris*, *Providencia rettgeri* – *Proteus rettgeri*, and *Morganella morganii* – *Proteus morganii*), *Pseudomonas aeruginosa*, *Staphylococcus aureus*, *S. epidermis*, and group D streptococci.

RECOMMENDED DOSAGE:

To be taken orally. Should be taken 1 hour before meal or 2 hours after meal with a glass of water.

Usual adult dose:

Urinary Tract infection: 400mg twice daily for 3 –10 days. Treatment may need to be continued for up to 12 weeks in chronic relapsing urinary-tract infections; it may be possible to reduce the dose to 400mg once daily if there is an adequate response within the first 4 weeks.

Dose may need to be reduced in renal impairment; 400mg once daily has been suggested when the creatinine clearance is 30ml or less per minute.

A single dose oral dose of 800mg has been given in the treatment of gonorrhoea.

ROUTE OF ADMINISTRATION:

Oral

CONTRAINDICATIONS:

Contraindicated if there is a history of hypersensitivity to norfloxacin or other quinolones.

WARNINGS AND PRECAUTIONS:

Warnings:

Exacerbation of myasthenia gravis

Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post marketing serious adverse events, including deaths and requirement for ventilator support has been associated with fluoroquinolones use in person with myasthenia gravis. Avoid fluoroquinolones in patients with known history of myasthenia gravis.

Precautions:

Use with caution in patients with epilepsy or a history of CNS disorders. Since norfloxacin and related fluoroquinolones have been shown to cause degenerative changes in weight-bearing joints of young animals, it has been suggested that these compounds should not be used in children, adolescents, pregnant women, or breast-feeding mothers. Care is necessary in patients with impaired hepatic or renal function, glucose-6- phosphate dehydrogenase deficiency, or myasthenia gravis. An

adequate fluid intake should be maintained during treatment and excessive alkalinity of the urine avoided because of the risk of crystalluria. Exposure to strong sunlight should also be avoided. The ability to drive or operate machinery may be impaired, especially alcohol is also taken. Should not be used in children and pregnant women. Phototoxicity may occur.

INTERACTIONS WITH OTHER MEDICAMENTS:

Norfloxacin may interact with various compounds including antacids, iron preparations, theophylline, and warfarin. It is recommended that norfloxacin should not be administered by mouth within 4 hours of taking preparation containing magnesium, aluminium or iron salts.

PREGNANCY AND LACTATION:

Should not be used in pregnant women and in lactating women.

SIDE EFFECTS:

Norfloxacin is generally well tolerated. The most often adverse effects involve the gastro-intestinal tract, central nervous system or skin.

Gastro-intestinal disturbances: Nausea, vomiting, diarrhoea, abdominal pain, and dyspepsia are the most frequent adverse effects. Pseudomembranous colitis has been reported rarely.

CNS: Headache, dizziness and restlessness are among the commonest effect reported. Others include tremor, drowsiness, insomnia, bad dreams, and visual disturbances and more rarely, hallucinations, psychotic reactions, depression and convulsions. Peripheral neuropathy has occurred occasionally.

Skin: In addition to rash and pruritus, hypersensitivity-type reactions affecting the skin have included photosensitivity and rarely, vasculitis, Stevens-Johnson syndrome, and toxic epidermal necrolysis. Anaphylaxis has been reported.

As with other quinolone antibacterial agents, reversible arthralgia has sometimes occurred, but joint erosions have only been documented in immature animals.

Exacerbation of myasthenia gravis

Post Marketing Experience.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Since there is no specific antidote for norfloxacin overdose, treatment should be symptomatic and supportive and may include the following:

- Induction of emesis or use of gastric lavage to empty the stomach.
- Maintenance of adequate hydration.
- Careful observation of the patient.

EFFECT ON ABILITY TO DRIVE AND USE MACHINE:

The ability to drive or operate machinery may be impaired.

PRECLINICAL SAFETY DATA :

Not applicable.

INSTRUCTION FOR USE :

To be taken orally, 1 hour before meal or 2 hours after meal.

STORAGE CONDITION :

Store below 30°C in well-closed, light resistant containers.

Protect from light and moisture.

Keep the medicine out of reach of children/ Jauhi dari kanak-kanak.

DOSAGE FORMS AND PACKAGING AVAILABLE:

Dosage form: Tablet

Packaging Available: Blister Pack: 100 x10's and 50 x 10's

NAME AND ADDRESS OF MANUFACTURER / PRODUCT REGISTRATION HOLDER:

MALAYSIAN PHARMACEUTICAL INDUSTRIES SDN BHD (101323-U)

Plot 14, Lebuhraya Kampung Jawa, 11900 Bayan Lepas,

Pulau Pinang, Malaysia.

DATE OF REVISION:

30/08/2018